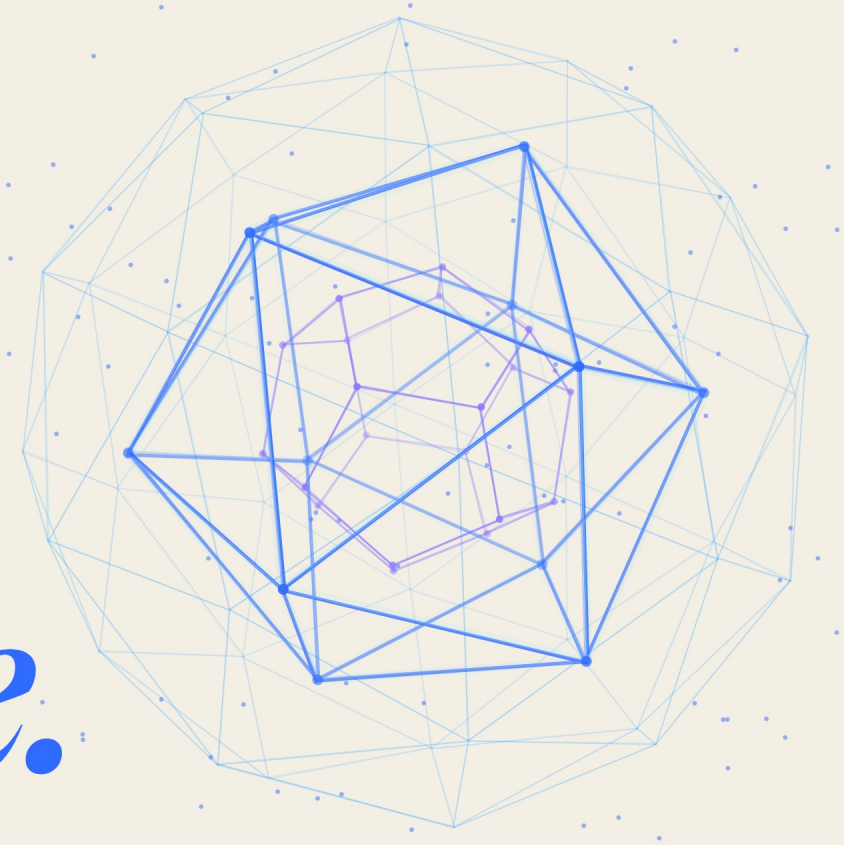
4ward

Pay later. At today's *price.*



Lock today's exchange rate for a supplier payment due later. On the due date, your supplier is paid in their own currency.

FX hedging for SMEs, built on Stellar

HackMeridian 2026 · Genesis

Between the invoice and the payment, the rate *moves*.

Large companies lock the rate with their bank. SMEs can’t, so the swing eats their margin.

≈3,000

USDC lost on a 1,000,000 MXN invoice when the peso makes an ordinary 90-day move.

≈34%

of imports in developing economies are made by SMEs.

Unhedged

Most SMEs use no hedge: no credit line, no FX desk, invoices too small.

Sources: 4ward simulation (one standard deviation over 90 days) · OECD, 2019 · Chudasama et al., SSRN · Neumann, 2019

Lock it. Cap it. *Forget it.*

On the other side, liquidity providers deposit USDC in the pool and earn the spread. Example: 1,000,000 MXN, 90 days, ±10%.

1 → 2 → 3 → 4

Quote

Locked price: 55,013 USDC for 1,000,000 MXN in 90 days.

Reflector rate + interest rates

Lock

Deposit 5,501 USDC (10%): your protection and your maximum loss.

Held by the Soroban contract

Settle

Peso up: the pool pays you the difference. Peso down: you pay it.

Oracle rate at maturity

Pay

The supplier receives exactly 1,000,000 MXN.

Stellar path payment

Protection and payment run on *the same rails*.

USDC

The price is locked, held and settled in dollars.

Reflector

FX rates published on-chain, read at maturity.

Path payment

Dollars become pesos on the way to the supplier.

Soroban

One contract for pricing, hedges and the vault.

Unlocked: path payment, USDC SAC, Soroban events · Composed: Reflector FX, Stellar Wallets Kit, Etherfuse yields

Nothing can *default*.

04 Safe by design

Capped protection

Your deposit is the most you can lose. No margin calls.

A pool that can’t go bust

It refuses any hedge it could not pay out in full.

The due-date rate

Nobody can pick a better moment to settle.

Settles by itself

A keeper settles each hedge right after its due date.

Early exit

Close before the due date, at the market price.

Only the owner signs

Nobody can open a hedge with your money.

Each rule is enforced by the contract and covered by tests.

A fair price, *not a bet.*

0.59%

of the invoice, all-in, for 90 days: today's rate adjusted for 90 days of interest, plus a 0.5% spread.

A classic currency option costs 2.2–2.6%.

If the peso jumps 12%, the extra cost over the locked price is:

Without protection



+6,587

With 4ward: only the part beyond your ±10% protection



+1,086

USDC, 1,000,000 MXN, 90 days, ±10%. Option: Garman–Kohlhagen, 11–13% vol. Source: 4ward simulation.

Built for *SMEs.*

	FX Swap by Hedgehog	4ward
For	DeFi users	SMEs paying foreign suppliers
Currencies	USDC, EURC, AUDD	Any oracle currency (MXN, BRL, ARS, EUR today)
Settlement	Physical delivery	In USDC, no currency stock
Supplier payment	Not included	Included, by path payment

FX Swap by Hedgehog: Stellar Community Fund #21, as described in its public materials.

The closest project trades currencies. *4ward pays the invoice.*

07 Potential

One dollar price, *any currency*.

4 → any

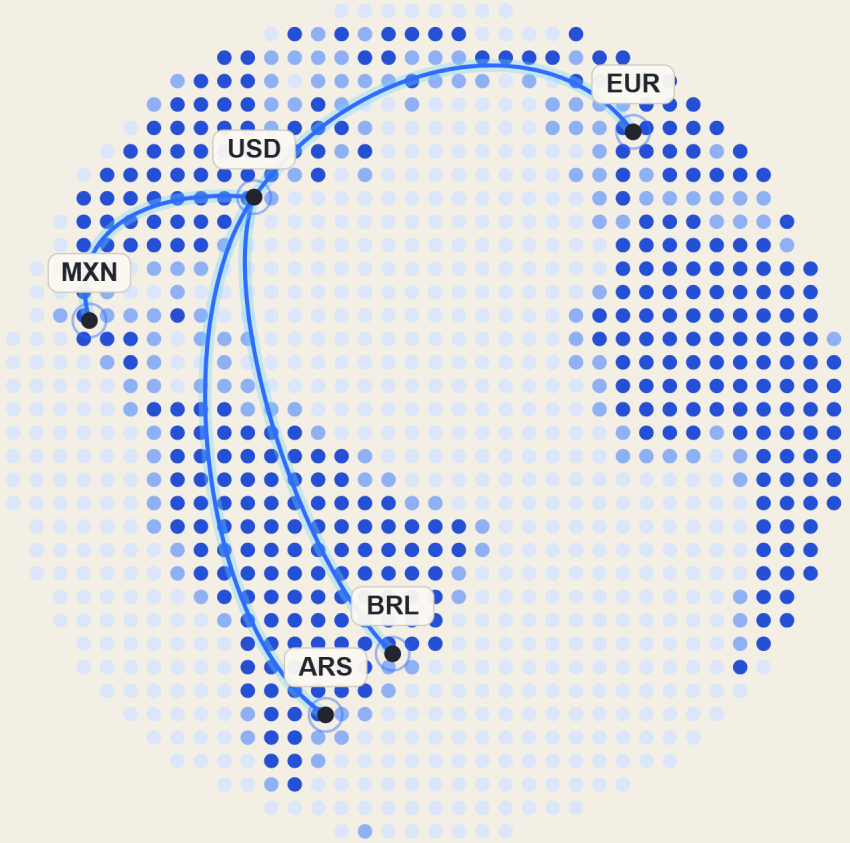
MXN, BRL, ARS and EUR today.

$210B

of SME imports a year in Mexico alone. Hedging just 0.1% of it means $210M a year.

1 SDK

“Lock the rate” inside the wallets and anchors where SMEs already pay.



Estimate: $626B Mexican imports (UN Comtrade, 2024) × 34% SME share (OECD average). Supplier payment shown in MXN on testnet.

From testnet to *the first invoices.*

Live on testnet

- Lock, settle at the due-date rate, exit early
- Two-tranche pool that can’t go bust
- Supplier paid by path payment
- Keeper for automatic settlement

Next

- Settle and pay in one atomic step
- Match importers with exporters
- Put the pool’s idle capital to work
- A “Lock the rate” SDK for wallets

Try it: 4ward-fx.vercel.app

The *team.*



Hugo Thomas

Vault & web app

Built the two-tranche vault and the whole app.



Lancelot Watelle

Pricing & fixing

Built the forward pricing and the maturity fixing.



Clément Darmon

Settlement & payment

Built settlement, the keeper and the supplier payment.

Three builders, *one working protocol*, 27 hours.